

# Hani Bounoua | Data Scientist / Data Mining Expert

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## Professional Summary

Data professional with deep expertise in the telecommunications sector, research, and freelance pharmacometrics. Proven track record of developing machine learning models, deploying interactive geo-analytics dashboards, and building robust large-scale data pipelines. Adept at transforming complex datasets into actionable business intelligence using advanced R and Python ecosystems.

## Technical Skills

**Programming & Databases:** Python, R, Teradata SQL, SQL Server, Java, C++

**Data Engineering & Big Data:** PySpark, Polars, pandas, Teradata (FastExport, FastLoad, Macros)

**Machine Learning & Modeling:** scikit-learn, tidymodels, mlr3, nlmixr2

**Web Apps, Viz & Tools:** R Shiny, Tableau, Power BI, Leaflet, Excel, NodeJS, ExpressJS

**DevOps & OS:** GitLab CI/CD, Git, Linux (Fedora, Garuda), Windows 11

## Professional Experience

### Reporting Administrator

February 2022 – Present

*Djezzy (Veon Group)*

*Algiers, Algeria*

- **Achievements:** Awarded Best Employee Q1 2026. Successfully passed technical and oral evaluations for the internal **Data Scientist Certified Expert** designation with a top score of 90.27% (official HR title update pending).
- **Predictive Modeling & Churn:** Developed machine learning models to detect rotational churn, predict inactivity within the 15-day tenure segment, and forecast the lifetime value of "bundler" clients based on 3-month behavioral histories. Built forecasting models for revenue segments.
- **Customer Segmentation & Analysis:** Automated Multiplay segmentation and conducted revenue cohort analysis based on behavioral segments. Analyzed the impact of market performance, deployment, and traffic indicators on YoY revenue by commune. Conducted post-monetization tracking for new services.
- **Network Optimization & Scoring:** Engineered a Regional Network Satisfaction Score (RNSS/CSS) and customer engagement scores using survey results and behavioral data. Proposed decision-support solutions to optimize boost campaigns and deployment processes based on capacities. Segmented network cells by traffic flow to detect congestion.
- **Dashboards & Reporting:** Created prototype commercial dashboards (Prepaid, Regional Performance, VAS, and Digital Services) using R Shiny. Automated complex reporting (B2B Revenue, In-flow/Outflow YoY, Server Refill Detection, Tenure Technology, Daily 5G performance) using Teradata SQL macros.

### Freelance Data Scientist & Pharmacometrician

December 2021 – Present

*Independent Consultant*

*Remote*

- **Medisim SaaS:** Architected and developed a modular Software as a Service (SaaS) application for processing, analyzing, and modeling pharmacokinetic data (with or without covariates), integrating a comprehensive reporting module.
- **SAA Deduplication:** Created a centralized unique identifier to consolidate databases for the Société Nationale d'Assurance (SAA) utilizing fuzzy string matching on insurance policyholder characteristics.

### Research Support Engineer

May 2020 – November 2021

*CERIST*

*Algiers, Algeria*

- Analyzed and reported on the overall performance of scientific research in Algeria leveraging data from Scopus and Web of Science databases.

- Developed a solver for a heterogeneous vehicle routing problem with time windows, pickup, and delivery constraints.

**Teaching Assistant / Instructor**

*Université M'Hamed Bougara de Boumerdès (UMBB)*

**December 2019 – February 2020**

*Boumerdès, Algeria*

- Taught practical and theoretical modules in R Programming (Tidyverse) and Numerical Analysis.

**Forecast and Objectives Officer**

*Condor Electronics*

**August 2019 – October 2019**

*Algeria*

- Managed databases for operational reporting and developed sales forecasting models to optimize inventory.

**Education**

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**Master's in Applied Statistics and Probability** | *USTHB, Algiers, Algeria*

**2016 – 2019**

**Bachelor's in Algebra and Cryptography** | *USTHB, Algiers, Algeria*

**2012 – 2016**

**Certifications**

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**Machine Learning & AI** (ITComp) | **Tableau 2022 Advanced** (Udemy)

**Advanced SQL Server for Data Analysis** (Udemy) | **Power BI Desktop** (IT Talents School)

**Languages & Interests**

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**Languages:** Arabic (Native), French (Bilingual), English (Professional Working Proficiency)

**Interests:** Running & Trail Running, Sciences & New Technologies